Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years translating customer requirements into **solution designs**, building **data integration pipelines**, **prototypes**, and analytics on **Google Cloud**, AWS, and Azure through close **stakeholder management**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Web Technologies, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Data Retrieval, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Designed the relational schema and storage architecture for a multi-tenant platform on Azure SQL and Blob Storage, enforcing **row-level data isolation** for 500+ users and reducing dashboard latency from 1.2s to 350ms.
- Built ETL-style **data integration pipelines** across FastAPI, Supabase, SQL Server, and cloud storage with incremental loads and reconciliation checks, processing 5K+ monthly jobs at 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Designed the end-to-end **solution flow** for an AI grading platform, from upload through rubric generation and feedback, integrating OpenAI, Anthropic, and Qwen models behind a switchable architecture for 150+ instructors.
- Tracked token usage and estimated cost for every model call across grading, rubric, and OCR workflows, enabling **cost-aware** model decisions while serving 10K+ daily requests and cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Partnered with Microsoft and GitHub **stakeholders** in Agile sprints, contributing from requirements analysis through deployment of cross-platform data-sharing microservices validated by 20+ unit and integration test suites.
- Consolidated user activity logs from multiple sources into unified formats with AWS Glue and S3 for **analytics**, improving pipeline efficiency 68% and response times 30% while cutting storage costs 15%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Applied **Google Analytics** feature-usage, traffic-source, and drop-off insights to guide CRM improvements, reducing resource wastage 42% and increasing lead conversions 15% on a MERN platform syncing 2M+ records.
- Defined boundaries for eight microservices covering authentication, resource allocation, billing, leads, permissions, and notifications, then automated Jenkins and Docker releases to staging with manual approval gates before production.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Designed the end-to-end booking system architecture for a client, mapped 12+ backend integration points, and **presented the solution** to the client for alignment, reducing design errors 25% before implementation began.
- Built an hourly Databricks and Spark **analytics pipeline** over Firebase booking and payment events, aggregating peak hours, top services, daily active users, and revenue summaries across 50K+ records for reporting.

Skills

Consulting & Delivery: Solution Design, System Design, Requirements Analysis, Stakeholder Management, Client Presentations, Cross-Functional Collaboration, Technical Documentation, Proof-of-Concept Prototyping, Agile/Scrum
Data & Analytics: Google Analytics, ETL Pipelines, Data Integration, Data Validation, Data Modeling, Databricks, Spark, SQL, PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB
Languages & Engineering: Java, Python, C++, JavaScript, TypeScript, REST APIs, Microservices, FastAPI, Flask, React.js
Cloud & AI: Google Cloud (GCP), AWS (S3, Glue, RDS), Microsoft Azure, Docker, Jenkins, CI/CD, Git, LLM Systems, FAISS

Projects

Big-Brain Live | *GCP, LLaMA-2, FAISS, Pinecone, PostgreSQL, REST APIs*

- Prototyped an AI document platform on **Google Cloud** with LLaMA-2 and FAISS semantic search, delivering 95% query accuracy across 50K+ documents for 200+ users and reducing document review time 60%.

Chat2SQL Code | *Java, TypeScript, SQL, PostgreSQL, Snowflake, Docker*

- Extended a **Java** and TypeScript SQL workspace supporting 10+ databases including PostgreSQL and Snowflake, adding schema-aware query generation plus AI-assisted reporting and dashboards from one unified interface.

HMIS - Hospital Management Information System Live | *TypeScript, MERN, ABDM APIs, Twilio, Sentry*

- Won Smart India Hackathon with a healthcare platform integrating ABDM APIs through explicit **patient-consent** flows, giving doctors 2x faster history access and managing 100+ daily appointments with Twilio reminders.